Competition between magnetic impurities and superconducting order in Nb thin films

Research Internship

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Abstract. This work investigates the competition between magnetic pair-breaking and superconducting order in 15 nm thick Niobium thin films patterned into Hall cross geometries. Three samples are studied: a pristine reference (G0Nb) and two sets doped with Mn^+ ions at densities of 10^{12} ions/cm^2 (G12Nb) and 10^{14} ions/cm^2 (G14Nb). All samples exhibit critical temperatures $T_c \approx 6$–7 K, well below the bulk Nb value of 9.2 K, attributed to thin-film disorder, surface oxidation, and reduced dimensionality. The IV characteristics display pronounced hysteresis between the switching current I_s and the retrapping current I_r, quantitatively accounted for by the Skocpol–Beasley–Tinkham (SBT) self-heating model. Furthermore, a focus has been made on the doped G12Nb sample. Mn^+ ion migration under a 150 V bias paradoxically decreases the normal-state resistance, interpreted as a redistribution of clustered interfacial impurities. This results in a T_c increase alongside a suppression of the onset T_c. This is confirmed by the $H_{c2}(T)$ phase diagram and by the emergence of multi-step vortex-flow features in the post-migration IV characteristics.

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Chapter 1

Context of the study

1.1 Work interest

The discovery of superconductivity by Kamerlingh Onnes in 1911, when mercury was found to lose all electrical resistance below 4 K, initiated one of the most fertile research topics in condensed-matter physics [1]. Since then, a lot of metals and alloys are found to be superconducting materials such as Al or Nb. Today, superconducting materials underpin technologies ranging from MRI magnets to superconducting single-photon detectors (SSPDs), which detect individual photons and outperform all competing technologies in low-light quantum-information applications.

A key challenge in device engineering is that superconductivity is fragile. It is easily destroyed by temperature, current, or magnetic field exceeding their respective critical values. It is further suppressed by disorder and, especially, by magnetic impurities. Magnetic atoms embedded in a superconducting host interact with the conduction electrons through an exchange coupling, causing spin-flip scattering that breaks Cooper pairs, which are pairs of electrons bound by electron-phonon coupling and forming the superconducting condensate. The pair-breaking effect is captured quantitatively by the Abrikosov–Gor'kov theory, which predicts a systematic decrease of the critical temperature T_c as the magnetic scattering rate increases.

The present work uses Niobium (Nb) thin films as a model system. Nb has the highest critical temperature (9.2 K) of any elemental superconductor at ambient pressure, is a definitive Type-II superconductor in thin-film form, and supports the native Nb_2O_5 oxide exploited in Josephson junctions. 15 nm thick films are patterned into Hall cross geometries and deliberately doped with Mn^+ ions to explore the competition between magnetic pair-breaking and superconducting order. The study is directly inspired by the thesis of Paul Amari (LPEM, Sorbonne Université, 2020), which demonstrated vortex-flow and switching-current physics in ion-irradiated YBCO superconducting nanowires, and which provides the theoretical and methodological framework applied here [2].

1.2 Theoretical Background

This section outlines the foundational theoretical frameworks of superconductivity relevant to our study, tracing the evolution from phenomenological descriptions to microscopic theory and the behavior of magnetic impurities.

1.2.1 Superconducting condensate description

The first successful description of superconducting electrodynamics was proposed by the London brothers in 1935. They introduced a *two-fluid* model where a fraction n_n of the electrons are normal electrons behaving according to Ohm's law, while the remainder n_s are part of the superconducting condensate and accelerate without resistance in an electric field $\vec{E}$:

$$\frac{\partial \vec{J}_s}{\partial t} = \frac{n_s e^2}{m}\vec{E} \qquad \text{and} \qquad \vec{J}_n = \sigma_n \vec{E} \tag{1.1}$$

When combined with Maxwell's equations, this leads to the second London equation $\nabla^2 \vec{B} = \vec{B}/\lambda_L^2$, which predicts the exponential decay of a magnetic field within the material. This characteristic length, known as the London penetration depth λ_L, describes how deep a field can penetrate before being screened by surface supercurrents:

$$\lambda_L(T) = \sqrt{\frac{mc^2}{4\pi e^2 n_s(T)}} = \frac{\lambda_0}{\sqrt{1-(T/T_c)^4}} \tag{1.2}$$

This mathematical framework explains the Meissner-Ochsenfeld effect, wherein a superconductor behaves as a perfect diamagnet by expelling all interior magnetic flux [3]. In an external field $\vec{H}$, the internal flux density is $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \vec{0}$, implying a perfect shielding magnetization $\vec{M} = -\vec{H}$.

The 1950 Ginzburg-Landau (GL) theory moved beyond electrodynamics to describe superconductivity as a second-order phase transition. It introduces a complex order parameter, $\psi(\vec{r}) = |\psi(\vec{r})|e^{i\varphi}$, where $|\psi|^2$ represents the local density of superconducting electrons. Minimizing the GL free energy functional reveals two fundamental length scales: the penetration depth λ_{GL} and the coherence length ξ. The latter represents the characteristic distance over which the order parameter varies. Depending on the material's purity (mean free path l_0), ξ is defined in two regimes:

$$\begin{cases} \xi(t) = 0.74\dfrac{\xi_0}{\sqrt{1-t}} & \text{in the clean limit } (l_0 \gg \xi_0) \\ \xi(t) = 0.86\dfrac{\sqrt{\xi_0 l_0}}{\sqrt{1-t}} & \text{in the dirty limit } (l_0 \ll \xi_0) \end{cases} \tag{1.3}$$

While GL theory is semi-phenomenological, the 1957 Bardeen-Cooper-Schrieffer (BCS) theory provides the microscopic origin [4]. It posits that an attractive interaction mediated by lattice vibrations (phonons) allows electrons with opposite momenta and spins to form Cooper pairs. This pairing creates an energy gap Δ in the electronic density of states, protecting the condensate from low-energy scattering. The Cooper pair extends over the coherence length $\xi_0 = \hbar v_F/\pi\Delta_0$. At $T = 0$, the gap is related to the critical temperature by $2\Delta_0 = 3.52 k_B T_c$.

1.2.2 Abrikosov vortices

The ratio of the two GL length scales, defined as the Ginzburg-Landau parameter $\kappa = \lambda/\xi$, determines the magnetic response of the material. Superconductors are categorized into two types:

- Type I ($\kappa < 1/\sqrt{2}$): These materials remain perfectly diamagnetic until a critical field H_c is reached:

$$H_c(T) = \frac{\Phi_0}{2\sqrt{2}\pi\mu_0\lambda(T)\xi(T)} \tag{1.4}$$

  Above this field, the external magnetic field is no longer screened, and the material reverts to its normal state.

- Type II ($\kappa > 1/\sqrt{2}$): As is the case with our Niobium (Nb) films, these materials exhibit a *mixed state* between two critical fields, H_{c1} and H_{c2}. The lower critical field H_{c1} marks the entry of the first Abrikosov vortex, which is a normal core of radius ξ carrying a single flux quantum $\Phi_0 = h/2e$, surrounded by circulating supercurrents. These vortices repel each other and typically organize into a triangular Abrikosov lattice. The critical fields for Type II superconductors are given by:

$$H_{c1}(T) = \frac{\Phi_0}{4\pi\mu_0\lambda^2(T)}(\ln\kappa - 0.18) \qquad H_{c2}(T) = \frac{\Phi_0}{2\pi\mu_0\xi^2(T)} \tag{1.5}$$

  At H_{c2}, the vortex cores overlap entirely, suppressing the order parameter and restoring the normal state.

1.2.3 Abrikosov-Gor'kov theory

Superconductivity is highly sensitive to defects that break time-reversal symmetry. While non-magnetic impurities shift the material into the *dirty limit*, magnetic impurities (such as Manganese ions) act as potent depairing centers. For s-wave superconductors like Niobium, magnetic ions cause spin-flip scattering that disrupts the singlet state of Cooper pairs. According to the Abrikosov-Gor'kov theory, the critical temperature T_c decreases as impurity concentration increases:

$$\ln\left(\frac{T_{c_0}}{T_c}\right) = \psi\left(\frac{1}{2} + \frac{\hbar}{4\pi k_B T_c \tau_s}\right) - \psi\left(\frac{1}{2}\right) \tag{1.6}$$

where ψ is the digamma function, T_{c_0} is the critical temperature of the pristine material, and τ_s is the spin-flip scattering time.

In the dilute limit, this theory predicts a linear initial suppression of T_c. Beyond a critical concentration (the depairing limit), superconductivity is entirely destroyed. This pair-breaking is also manifested experimentally as a broadening of the resistive transition and an enhancement of normal-state resistivity, in accordance with Matthiessen's rule.

Chapter 2

Device fabrication

2.1 Considered devices

The primary objective of this work is to characterize Niobium (Nb) thin films with 15 nm thickness, patterned into Hall cross geometries and doped with Mn^{+} ion impurities. While high-purity bulk Niobium exhibits a critical temperature $T_c \approx 9.2$ K and a BCS coherence length $\xi_0 \approx 38$ nm, our films operate in the *dirty limit* due to their reduced dimensionality and structural constraints. In this regime, the effective coherence length ξ is suppressed to approximately 10–20 nm.

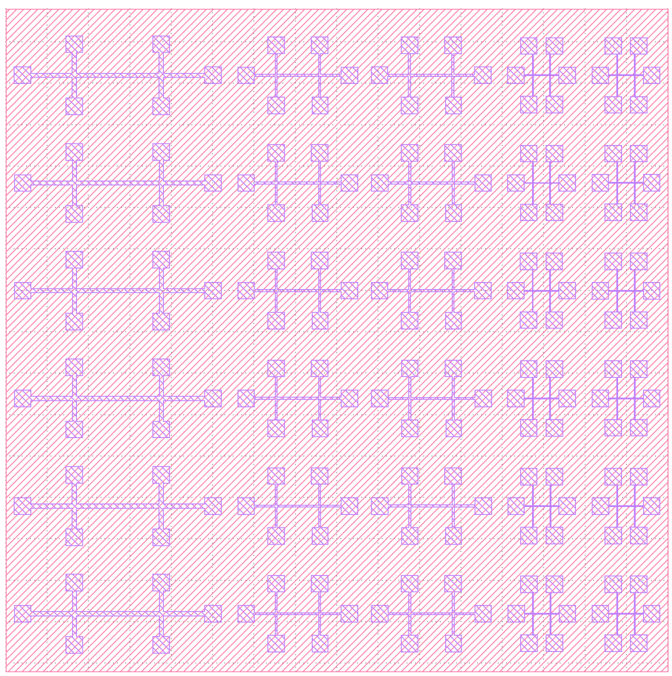

Figure 2.1 – KLayout design of the Hall crosses etched via photolithography (in purple). The central channel widths w vary (50 μm, 25 μm (x2), and 10 μm (x2)) with a constant length-to-width ratio $L = 20w$. Contact pads are 200 μm $\times$ 200 μm squares. A minimum distance of $5w$ is maintained between the pads and the junction to ensure uniform current distribution across the channel.

Given that the London penetration depth λ_L in Nb is typically comparable to ξ_0, bulk Niobium resides near the boundary between Type-I and Type-II superconductivity. However, in thin films, the reduced electronic mean free path significantly increases the Ginzburg-Landau parameter $\kappa = \lambda/\xi$, ensuring that our samples behave as definitive Type-II superconductors.

Niobium was selected as the material of choice for several reasons:

- High transition temperature: Nb has the highest T_c (9.2 K) of any elemental superconductor at ambient pressure, facilitating stable measurements using liquid Helium cooling.

- Magnetic robustness: As a Type-II superconductor, Nb maintains its superconducting state under significantly higher magnetic fields than Type-I materials.
- Chemical stability: The formation of a stable, thin native oxide (Nb_2O_5) provides a protective barrier, making Nb the industry standard for superconducting circuits and Josephson junctions.

The study focuses on three samples with varying magnetic impurity concentrations to investigate the pair-breaking effects described by the Abrikosov-Gor'kov theory:

Sample	$\mathbf{Mn^+}$ **(ions/cm^2)**	**Condition**
G0Nb	0	Pristine control
G12Nb	10^{12}	Dilute doping
G14Nb	10^{14}	High doping

Table 2.1 – Magnetic impurity densities across the considered samples.

2.2 Experimental procedure

2.2.1 Thin film deposition

Soda-lime glass substrates were cleaned via sequential sonication in acetone, isopropanol (IPA), and deionized water (3 minutes each). To introduce magnetic impurities, Mn^+ ions were implanted onto the cleaned glass surfaces using Reactive Ion Etching (RIE). The implantation was performed under an O_2 flow of 50 sccm at a power of 100 W for 1 minute ($P = 10^{-2}$ mbar).

The 15 nm Niobium layer was subsequently deposited via DC magnetron sputtering. Before deposition, an *in-situ* plasma cleaning (pickling) was performed under Argon flow (50 sccm, 50 W, 1 minute, RF) at 1.1 Pa. The Nb target was pre-sputtered for 2 minutes at 600 W to remove surface contaminants. Finally, the film was deposited at 19°C with the following parameters:

- Gas environment: Ar at 50 sccm (1.1 Pa working pressure).
- Deposition power: 500 W (RF) for 10 seconds.
- Base pressure: 2.3×10^{-7} mbar.

2.2.2 Patterning via photolithography

To study the influence of lateral confinement on superconducting transport, we fabricated Hall crosses with varying widths w (10 μm, 25 μm, and 50 μm). The device patterning followed a standard cleanroom workflow:

1. Surface pre-treatment: Following a standard solvent cleaning, a dehydration bake was performed at 120°C for 4 minutes to enhance resist adhesion.
2. Spin coating and exposure: A positive photoresist was spin-coated at 4000 rpm for 40 seconds. The pattern was defined via UV exposure through a chrome-on-glass photomask.
3. Development: The latent image was developed in a Metal-Ion Free (MIF) developer, dissolving the UV-exposed regions of the resist.
4. Etching: The final structure was etched into the Nb film using RIE. The remaining photoresist acted as a sacrificial mask, protecting the Hall cross geometry while the exposed Niobium was removed.

Chapter 3

Superconducting phase characterization

3.1 Measurement configuration

Initial transport characterization was performed using a four-terminal (Kelvin) sensing configuration. This geometry is essential for isolating the intrinsic resistance of the Niobium thin films by eliminating the contribution of parasitic lead and contact resistances. Measurements were conducted in a Kapitsa-type cryostat, with sample temperatures regulated by a LakeShore PID controller.

DC excitation was provided by a Yokogawa GS200 high-precision current source, while the resulting longitudinal voltage drops were recorded using a Keithley 2182A nanovoltmeter. For measurements requiring sub-microvolt resolution or the rejection of high-frequency noise, Stanford Research Systems (SRS) lock-in amplifiers were utilized. To ensure thermal equilibrium, samples were mounted onto a high-thermal-conductivity oxygen-free copper holder using silver paste, providing robust thermal anchoring between the film and the cryostat's heat sink.

3.2 Transport characterization

To maintain experimental consistency, all subsequent analysis focuses on devices with a channel width of $w = 50\ \mu$m.

3.2.1 Resistance–Temperature (RT) characteristics

The RT characteristics were recorded to establish the baseline superconducting properties of the films. The measurements were performed with a bias current $I = 1\ \mu$A and an RF environment of 100 MHz at -10 dBm to suppress environmental noise.

As the temperature drops below the critical temperature T_c, a phonon-mediated attraction overcomes Coulomb repulsion, causing electrons to pair up into Cooper pairs. The drop occurs as these pairs lock into a single, coherent quantum wave function. As long as the thermal and kinetic energy are below the superconducting energy gap (2Δ), the condensate flows with zero dissipation, moving through the lattice and around impurities without losing energy.

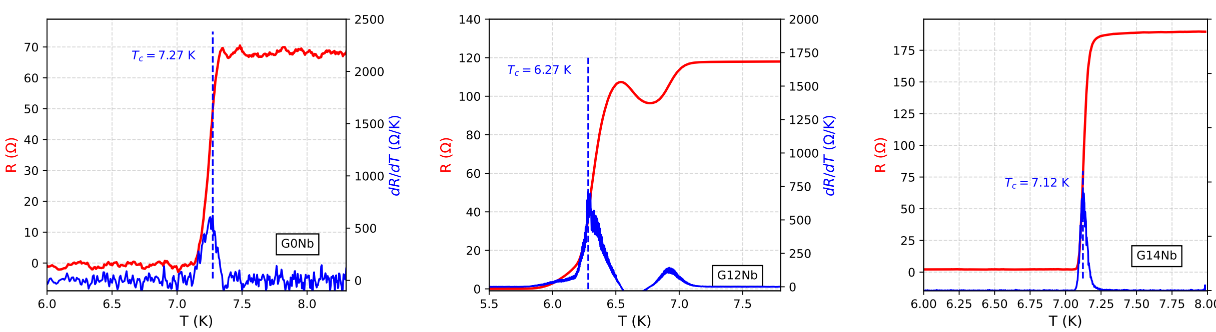

Figure 3.1 – RT characteristics and corresponding dR/dT derivatives. G0Nb exhibits the highest T_c (7.27 K) and lowest R_N. G14Nb shows a higher R_N but a remarkably sharp transition. G12Nb displays a suppressed T_c (6.27 K) and a broadened, multi-peak derivative, indicating phase inhomogeneity or contact-related artifacts.

The critical temperature T_c, is defined here as the temperature corresponding to the inflection point of the transition (the maximum of the derivative dR/dT), where approximately half of the normal-state resistance R_N is recovered. The transition width, ΔT_c, is defined as the temperature range between 10% and 90% of R_N. This width serves as a sensitive probe of the film's phase homogeneity. A sharp transition indicates a uniform superconducting gap, while a broad or multi-peaked transition suggests structural inhomogeneities or varying local T_c values.

Given that the film thickness ($d = 15$ nm) is significantly smaller than the bulk coherence length ($\xi_0 \approx 38$ nm), the system resides in the thin-film (quasi-2D) limit. In this regime, the normal-state resistivity ρ is a direct probe of electronic scattering:

$$\rho = R_N \frac{w \cdot d}{L} \tag{3.1}$$

where $L = 20w = 1$ mm is the channel length. The extracted parameters are summarized in Table 3.1.

Sample	R_N **(Ω)**	ρ **(μΩ · cm)**	T_c **(K)**	ΔT_c **(K)**
G0Nb	70.30	6.21	7.27	0.27
G12Nb	118.06	8.85	6.27	0.55
G14Nb	190.49	14.29	7.12	0.12

Table 3.1 – Superconducting transport parameters for the 50 μm Nb channels.

The measured T_c values (≈ 7 K) are lower than the bulk Niobium value (9.2 K), likely due to surface oxidation and structural disorder inherent to 15 nm films. Interestingly, G14Nb shows the highest resistivity but the sharpest transition ($\Delta T_c = 0.12$ K), suggesting a high density of uniform point defects. In contrast, G12Nb's broad transition and lower T_c indicate a less homogeneous superconducting state.

3.2.2 Current–Voltage (IV) characteristics

The IV characteristics provide insight into the dissipation mechanisms and critical current densities. In the superconducting state, a current density $j = I/(w \cdot d)$ flows with zero measurable voltage until a critical threshold is reached. We identify two primary regimes of dissipation:

- Flux creep: At low voltages, dissipation arises from thermally activated vortex hopping between pinning sites.
- Flux flow: At higher biases, the Lorentz force overcomes the pinning potential, leading to a linear regime where the differential resistance approaches the normal-state value.

The critical current evolution with temperature is clear: the more you decrease the temperature T below T_c, the more robust the superconducting state becomes, the higher the critical current. We also observe that the IV curves exhibit pronounced hysteresis, defined by the switching current I_s, where the film enters the resistive state, and the retrapping current I_r, where superconductivity is restored. Following the Skocpol-Beasley-Tinkham (SBT) model, this hysteresis is attributed to local Joule self-heating. Once a resistive *hotspot* forms at I_s, the local temperature exceeds the bath temperature, requiring the current to be lowered significantly ($I_r < I_s$) before the heat can be dissipated and the superconducting state recovered.

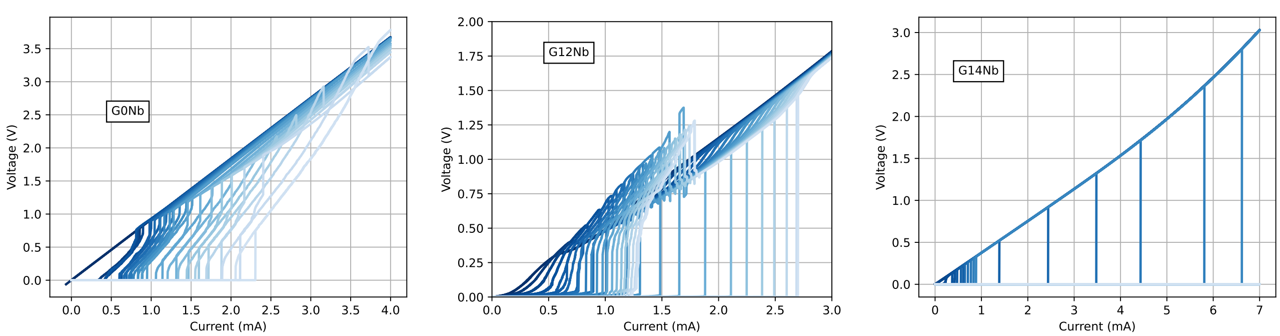

Figure 3.2 – IV characteristics for different operating temperatures. G0Nb : from 7.5 K (dark) to 4.5 K (light) with 0.15 K step. G12Nb : from 6.5 K (dark) to 4.6 K (light) with 0.1 K step. G14Nb : from 7.4 K (dark) to 4.5 K (light) with 0.1 K step.

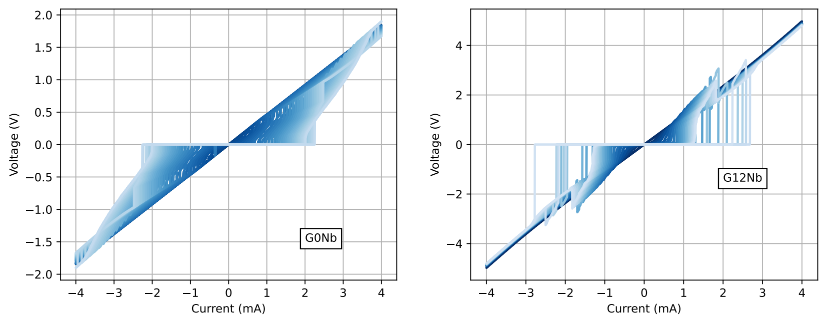

Figure 3.3 – IV characteristics with hysteresis checks for different operating temperatures. G0Nb : from 8.0 K (dark) to 4.5 K (light) with 0.1 K step. G12Nb : from 6.5 K (dark) to 4.6 K (light) with 0.1 K step.

The temperature dependence of the critical current density $j_c(T)$ was analyzed using several theoretical models (where $t = T/T_c$) [2]:

1. Bardeen formula in the dirty limit: $j_c(t) = j_c(0)(1 - t^2)^{3/2}$
2. GL depairing limit: $j_d(t) = j_d(0)(1 - t^2)^{3/2}(1 + t^2)^{1/2}$
3. Pinning models:
   - δl-pinning (Fluctuations in mean free path): $j_c(t) \propto (1 - t^2)^{5/2}(1 + t^2)^{-1/2}$
   - δT_c-pinning (Fluctuations in T_c): $j_c(t) \propto (1 - t^2)^{7/6}(1 + t^2)^{5/6}$

Sample	$j_r(0)$	$j_s(0)$	T_c^r	T_c^s	T_c^{RT}	**Best j_s model**
G0Nb	0.368	0.411	7.74	7.71	7.27	δT_c-pinning
G12Nb	0.259	0.973	6.95	6.81	6.27	Bardeen
G14Nb	0.314	23.96	7.40	7.40	7.12	δl-pinning

Table 3.2 – Fitting parameters extracted from IV characteristics (j in MA/cm^2, T in K).

The retrapping current j_r for all samples is best described by the δT_c-pinning model. This suggests that the return to the superconducting state is universally governed by the thermal recovery of regions with the lowest local critical temperatures.

For the switching current j_s, the limiting mechanisms diverge:

- G14Nb exhibits a massive $j_s(0)$ and follows the δl-pinning model. This suggests that the high dose of Mn ions creates a dense network of strong pinning centers related to mean-free-path fluctuations, which effectively *lock* vortices in place until very high current densities are reached.
- G12Nb follows the Bardeen model, indicating that its critical current is limited by the fundamental depairing of Cooper pairs in a disordered (dirty) medium, consistent with the phase inhomogeneity observed in *RT* measurements.

Finally, we observe that the critical temperatures derived from IV fits (T_c^s, T_c^r) are consistently higher than the T_c^{RT} measured at low bias. This discrepancy is a hallmark of the SBT model: while *RT* measurements are performed in a nearly isothermal state, the IV fits must account for the localized overheating of the electronic system. The higher extrapolated T_c reflects the theoretical transition temperature of the *hot* electronic population within the material.

Chapter 4

Influence of magnetic impurities

To investigate the impact of magnetic impurities on the superconducting condensate, sample G12Nb (10^{12} Mn^+ ions/cm^2) was studied in two distinct states: an *initial* state (post-implantation but pre-migration) and a *doped* state. The latter was achieved by applying a 150 V bias to induce a uniform migration of Mn^+ ions from the substrate-film interface into the bulk of the 15 nm Niobium layer.

4.1 Resistance–Temperature (RT) characteristics

4.1.1 Phase transition

The RT characteristics demonstrate that magnetic impurities fundamentally alter the nature of the superconducting transition. In pristine Nb, the transition is driven by the collective condensation of spin-singlet Cooper pairs ($\uparrow\downarrow$). However, Mn^+ ions have localized magnetic moments that act as magnetic pair-breakers. According to the Abrikosov-Gor'kov theory, these moments break time-reversal symmetry, leading to spin-flip scattering that disrupts the phase coherence of the pairs and suppresses the order parameter ψ.

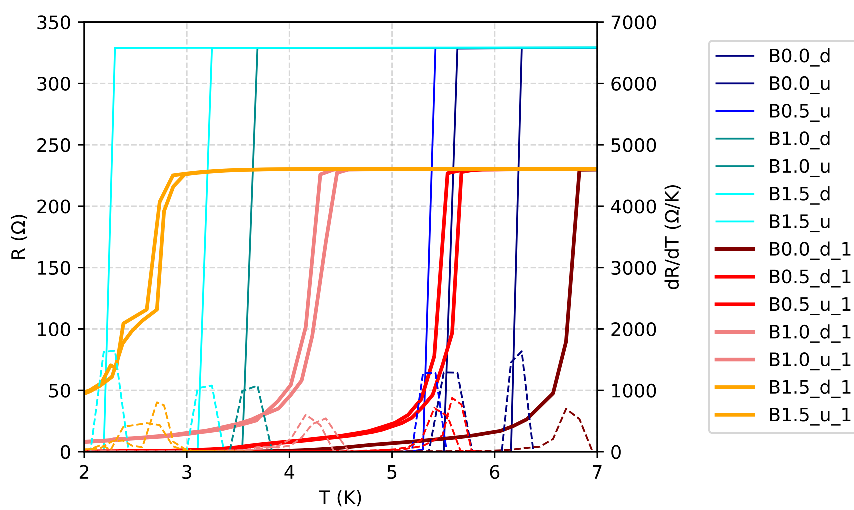

Figure 4.1 – Temperature dependence of resistance R and dR/dT (dashed plots) under magnetic fields (0 T, 0.5 T, 1 T, 1.5 T). In the caption, d and u correspond to the down or up orientation of the magnetic field, and _1 marks that the sample is doped. The broadened transitions (ΔT_c up to 0.88 K) and the emergence of resistive tails at high fields are indicative of magnetic pair-breaking and thermally activated vortex creep facilitated by the Mn^+ impurities.

The experimental RT characteristics show a significant increase in the transition width ($\Delta T_c \approx 0.5$–0.9 K) for the doped samples compared to the pristine Nb film. While a pure superconductor transitions abruptly, the presence of Mn^+ impurities introduces several physical mechanisms that modify this phase change:

- Spatial inhomogeneity: At the 15 nm scale, ion implantation and subsequent migration do not result in a perfectly uniform distribution of impurities. This creates a landscape of varying local critical temperatures $T_c(\vec{r})$. As the sample is cooled, *clean* regions with lower impurity concentrations reach their local T_c first and become superconducting islands. Because these islands are initially surrounded by a resistive normal-state matrix, the global resistance remains non-zero. The resistance only drops to zero when the temperature is low enough for the superconducting coherence length $\xi(T)$ to bridge the gaps between these islands, forming a continuous superconducting path across the Hall cross.

- Magnetic pair-breaking mechanism: The Mn^+ ions have localized magnetic moments that act as potent pair-breakers. When a Cooper pair electron scatters off a magnetic impurity, its spin orientation may be flipped. This disrupts the time-reversal symmetry required for the singlet state. Near T_c, the binding energy of the pairs is extremely small. The magnetic interference prevents the condensate from establishing long-range phase coherence, resulting in a *resistive tail* where superconductivity exists locally but cannot yet support a non-dissipative global current.

- Thermally-activated vortex creep: In Type-II thin films, the transition is sensitive to the motion of magnetic flux. Magnetic impurities create a complex landscape with varying potential barrier heights. At temperatures just below the onset of superconductivity, thermal energy ($k_B T$) is sufficient to allow vortices to *hop* or *creep* between pinning sites. This motion induces a voltage via the Josephson relation ($V \propto \partial\varphi/\partial t$), leading to a finite resistance that persists until the temperature is low enough to *freeze* the vortex lattice into a stationary state.

The broadening ΔT_c is therefore a macroscopic manifestation of the microscopic competition between the ordering force of the superconducting condensate and the disordering force of magnetic spin-flip scattering and thermal fluctuations.

B **(T)**	T_c^d **(K)**	T_c^u **(K)**	T_c^d **(K) (doped)**	**Doped** T_c^u **(K) (doped)**
0.0	5.5042	6.2639	6.6962	*nan*
0.5	*nan*	5.3020	5.4119	5.5858
1.0	3.6884	4.7015	4.1592	4.3549
1.5	2.2996	3.2446	2.6092	2.7087

Table 4.1 – Critical temperatures extracted from *RT* characteristics.

The *nan* values at 1.5 T suggest that the magnetic field successfully suppresses the transition below the measurable threshold of 2 K, or that the high disorder leads to a resistive tail that persists across the measured temperature range.

4.1.2 Resistivity anomaly

A counter-intuitive observation in the G12Nb sample is the evolution of the normal-state resistance R_N following the ion migration process. According to Matthiessen's rule, the introduction of Mn^+ ions into the 15 nm Nb bulk should enhance the scattering rate and increase R_N. However, experimental results demonstrate a significant decrease from $R_N \approx 330\ \Omega$ (undoped) to $R_N \approx 230\ \Omega$ (doped).

This suggests that the undoped state was not truly pristine, but rather characterized by a highly disordered inhomogeneous accumulation of impurities at the film-substrate interface. These clusters acted as high-resistance bottlenecks. The 150 V migration bias likely dissociated these clusters and redistributed the ions more uniformly throughout the layer. By *healing* these interfacial scattering centers, the effective electronic mean free path (l) increased, shifting the system toward the clean limit.

The physical interpretation of the transition depends critically on the chosen T_c criterion:

- Mid-point T_c ($50\% R_N$): This metric probes the *bulk* of the film. The doped sample shows a higher mid-point T_c, reflecting the overall reduction in structural disorder and the improved clean limit of the conduction channel.

- Onset T_c ($R \to 0$): This metric probes the *connectivity* of the superconducting condensate. In the doped state, this T_c is suppressed. While the bulk is cleaner, the now-distributed Mn^+ ions act as omnipresent magnetic pair-breakers that disrupt long-range phase coherence, preventing the formation of a zero-resistance path until lower temperatures.

4.1.3 Magnetic hysteresis

The RT characteristics exhibit a clear asymmetry between upward (u) and downward (d) magnetic field orientations. This phenomenon arises from the interplay between the cryostat's magnetic environment and the internal flux dynamics of the film. In fact, superconducting magnets exhibit magnetic hysteresis. A small positive remnant field (B_{rem}) typically persists even when the nominal current is zero. This field adds a static offset to the applied field, making the absolute magnitude $|B_{\text{applied}} \pm B_{\text{rem}}|$ intrinsically different for u and d sweeps.

Also, as a Type-II superconductor, the Nb film traps quantized Abrikosov vortices at Mn^+ pinning sites. When the field orientation is reversed, the existing population of *up* vortices must be annihilated by the ingress of *down* vortices. This process is dissipative and hysteretic, leading to different resistive profiles depending on the sample's magnetic history.

Finally, the Mn^+ ions introduce a paramagnetic or weak ferromagnetic response. Any structural non-uniformity or magnetocrystalline anisotropy within the 15 nm film could lead to an orientation-dependent pair-breaking effect, further discriminating the u and d curves.

4.1.4 Critical field (H_{c2}) analysis

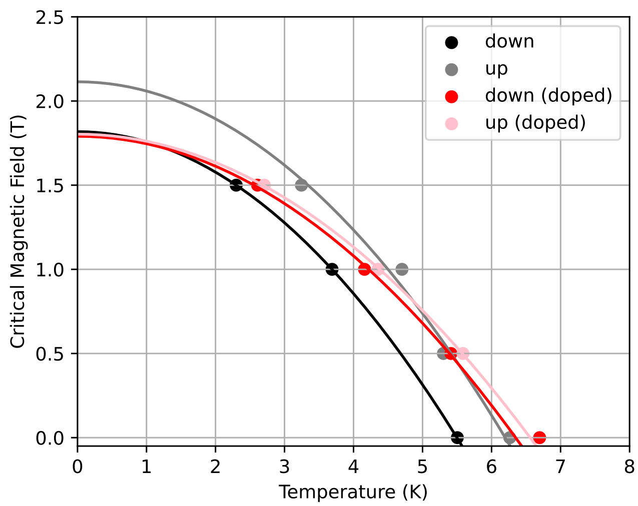

Figure 4.2 – Temperature dependence of the critical magnetic field. There are two distinct phases: the superconducting state below the curves and the normal state above them.

The $H_{c2}(T)$ phase diagram for sample G12Nb provides a macroscopic view of how ion migration and external magnetic fields modulate the superconducting order parameter. The data points were extracted using a resistance threshold (e.g., $50\% R_N$) and fitted using the Ginzburg-Landau (GL) relation:

$$H_{c2}(T) = H_{c2}(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right] \tag{4.1}$$

The most prominent feature of the graph is the significant rightward shift of the phase boundary following the 150 V ion migration process.

- Enhancement of T_c: In the *downward* field configuration, T_c increases from approximately 5.50 K (black curve) to 6.35 K (red curve). This supports the hypothesis that the migration bias redistributed Mn^+ impurities away from clustered interface sites, *cleaning* the conduction channel and reducing the normal resistance R_N.

- Evolution of $H_{c2}(0)$: Despite the increase in T_c, the extrapolated upper critical field at zero temperature $H_{c2}(0)$, appears relatively stable or slightly suppressed in the doped state. This reflects the competing influences of a cleaned conduction channel increasing T_c and distributed pair-breaking.

The separation between the *upward* (u) and *downward* (d) curves for both states highlights the influence of magnetic history and experimental environment. Specifically, the divergent paths of the phase boundaries reflect the Type-II nature of the Niobium film. The presence of Mn^{+} ions creates a landscape of pinning centers that trap Abrikosov vortices. The energy required to annihilate or rearrange these trapped vortices when reversing field direction results in the observed hysteretic behavior in the phase diagram.

4.2 Current-Voltage (IV) characteristics

4.2.1 Raw data analysis

The current-induced transition from the superconducting to the normal state in sample G12Nb reveals a fundamental change in the dissipative dynamics following ion migration.

- Emergence of intermediate states: While the undoped sample shows sharp switching, the doped sample exhibits complex IV shoulders. These features are characteristic of a vortex-flow regime, suggesting that Mn^{+} ions act as pinning centers and create a landscape of locally varying critical current densities.
- Phase inhomogeneity: The tiered structure of the RI characteristics in the doped state indicates a highly inhomogeneous superconducting order parameter. This inhomogeneity explains why the T_c defined by the $50\%R_N$ threshold increases, while the $R \neq 0$ onset remains sensitive to local magnetic pair-breaking.

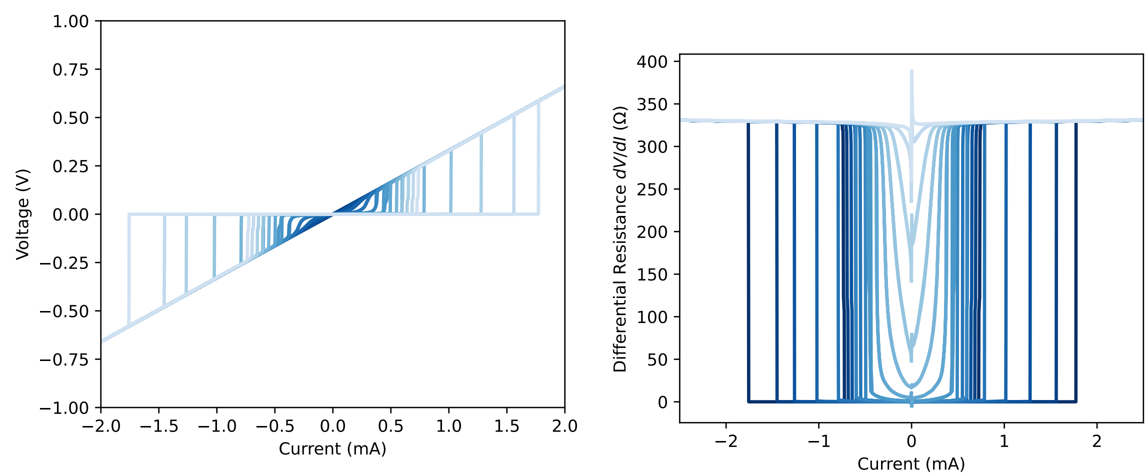

Figure 4.3 – IV characteristics and their corresponding RI characteristics for the undoped sample. Temperature varies from 7.1 K (dark) to 5.9 K (light) with 0.1 K step.

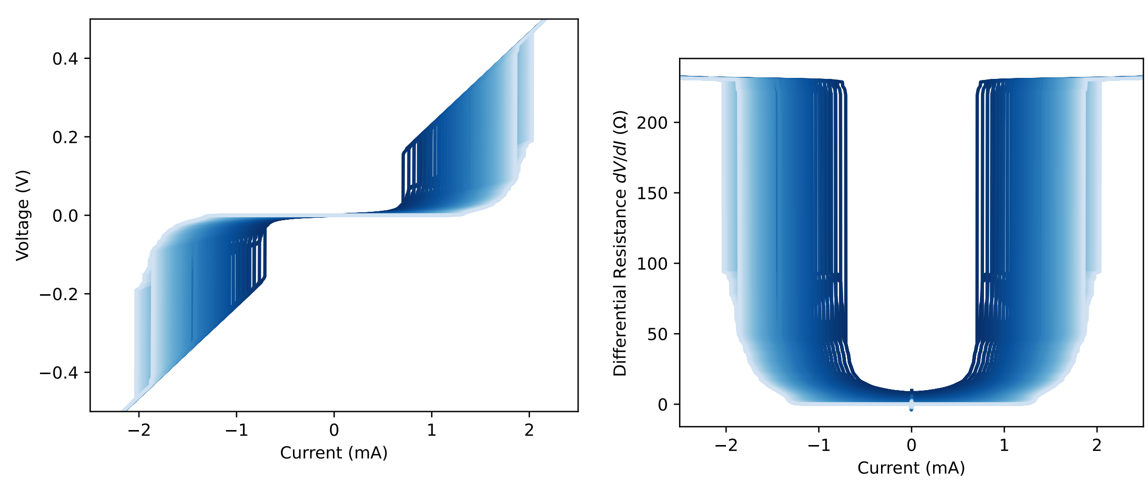

Figure 4.4 – IV characteristics and their corresponding RI characteristics for the doped sample. Temperature varies from 7.3 K (dark) to 2.0 K (light) with 0.1 K step.

4.2.2 Critical current density evolution

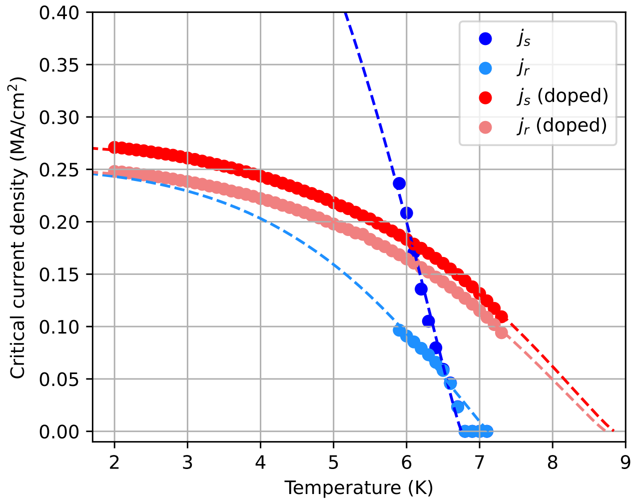

Figure 4.5 – Temperature-dependence of the critical current density (switching and retrapping) for the undoped and doped samples. Dashed plots are fitting models following the δT_c-pinning model.

Sample	$j_r(0)$ (MA/cm^2)	$j_s(0)$ (MA/cm^2)	T_c^r (K)	T_c^s (K)	T_c^{RT} (K)	j_r model	j_s model
G12Nb	0.251	0.758	7.13	6.75	6.27	δT_c-pinning	δT_c-pinning
Doped G12Nb	0.251	0.254	8.73	8.85	7.12	δT_c-pinning	δT_c-pinning

Table 4.2 – Extracted parameters from IV characteristic fits using the least squares method.

The most prominent feature we can notice is once again the rightward shift of the phase boundary. The extrapolated critical temperature (where $j \to 0$) increases from approximately 7.1 K in the undoped state to nearly 8.8 K in the doped state. This confirms that the 150 V bias successfully redistributed Mn^+ clusters away from high-disorder interface regions, effectively *healing* the conduction channel and allowing the film to behave more like bulk Niobium.

In the undoped state (blue), a large gap exists between j_s and j_r, accompanied by a very steep slope for the switching current. This suggests a high susceptibility to Joule self-heating and the formation of unstable hotspots. In the doped state (red), the j_s and j_r curves are significantly closer. This indicates that the uniform redistribution of impurities provides a more stable pinning landscape, which suppresses the thermal instability responsible for wide hysteresis.

Finally, as $T \to 2$ K, the critical current in the doped state saturates near 0.27 MA/cm^2. While the migration process significantly expanded the operational temperature range (T_c enhancement), the absolute magnitude of j_s remains constrained.

What's clear is that the $j(T)$ data provides conclusive evidence for the ion migration hypothesis. The electrical bias successfully optimized the device by shifting the impurity profile, resulting in a more robust superconducting state with improved thermal stability and a 25% increase in operational T_c.

Conclusion

This work investigated the interplay between magnetic impurities and superconducting order in 15 nm Niobium thin films using transport measurements at cryogenic temperatures. The principal findings can be summarised as follows.

All three Nb samples (G0Nb, G12Nb, G14Nb) exhibit critical temperatures near 7 K, well below the bulk Nb value of 9.2 K. This suppression reflects the combined effects of thin-film disorder, surface oxidation, and quasi-two-dimensional confinement. The IV hysteresis observed across all samples is consistently explained by the SBT thermal model: the retrapping current, governed by local δT_c fluctuations, is universal across samples, while the switching current mechanism is sample-specific — Bardeen depairing for the disordered G12Nb, and δl-pinning for the structurally uniform but highly resistive G14Nb.

The study of magnetic impurities in G12Nb revealed a richer phenomenology. The 150 V migration bias redistributed Mn^{+} ions from a clustered interfacial configuration to a more uniform distribution throughout the Nb film. This produced a paradoxical decrease in normal-state resistance, explainable as a healing of high-resistance interface regions. The simultaneous increase in midpoint T_c and decrease in onset T_c captures the dual nature of this redistribution: a cleaner bulk coexists with omnipresent pair-breaking centers that suppress long-range phase coherence. Under applied magnetic fields, the transition broadens substantially (ΔT_c up to 0.88 K) and resistive vortex-creep tails emerge, consistent with a thermally activated vortex landscape pinned by the Mn^{+} impurities. The $H_{c2}(T)$ phase boundary, fitted by the GL relation, shifts rightward after migration with a relatively stable $H_{c2}(0) \approx 1.80$–2.11 T, reflecting the competition between interface healing and distributed pair-breaking. Finally, the post-migration IV characteristics transition from sharp switching to multi-step vortex-flow shoulders, direct evidence for a spatially inhomogeneous critical current density landscape.

Several directions offer natural extensions of this work. First, the microscopic counterpart of the macroscopic pair-breaking observed here could be accessed through Yu–Shiba–Rusinov (YSR) bound states. Individual magnetic impurities in an s-wave superconductor induce sub-gap bound states within the superconducting gap, whose energy depends on the exchange coupling strength. Scanning tunnelling spectroscopy (STS) on single Mn atoms deposited on a Nb surface would allow direct imaging of these Shiba states and their spatial extent, connecting the single-impurity quantum level structure to the collective pair-breaking measured here — a strategy pursued in the context of quantum dot transistors by García Corral (Université Grenoble Alpes, 2020) [5]. Second, reducing the channel width toward the coherence length $\xi \approx 10$–20 nm would enter the one-dimensional regime where vortex crossings are replaced by phase slips as the primary dissipation mechanism, enabling direct comparison with the switching-current statistics of YBCO nanowires characterised by Amari (2020) [2]. Third, a systematic study of samples at intermediate Mn^{+} doses would allow a quantitative mapping of the Abrikosov–Gor'kov pair-breaking curve $T_c(n_s)$, providing a stringent microscopic test of the theory in a controlled thin-film geometry. Fourth, applying the thermally activated vortex motion (TAVM) framework to the low-voltage flux-creep regime of our IV curves would give access to the Pearl penetration length $\Lambda(T)$ and its evolution with doping, which is a direct measure of how pair-breaking modifies the two-dimensional superfluid density.

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